Topology versus Geometry in Single-Cell Trajectories: A Demonstration using Bonsai

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Abstract

Understanding cellular differentiation trajectories requires distinguishing between geometric similarity in embedded representations and the underlying lineage topology of cell states. Whilst dimensionality reduction methods such as PCA and UMAP are widely used to visualize high-dimensional single-cell data, they primarily preserve geometric relationships and do not explicitly recover ancestral structure. In this small demonstrative project we explore the recently proposed Bonsai algorithm for lineage inference using a synthetic branching trajectory dataset. The synthetic data provide a known ground truth trajectory embedded in a higher-dimensional feature space. We compare the inferred Bonsai lineage tree with PCA and UMAP embeddings to illustrate how geometric projections relate to lineage topology. The results show that although PCA partially separates the branches of the synthetic trajectory, the embedding alone does not reveal the underlying ancestral relationships. In contrast, the Bonsai method successfully reconstructs a branching lineage structure consistent with the ground truth.

1 Motivation

Recent work in systems biology has emphasized that biological systems can often be understood through the geometry of the high-dimensional state spaces in which they operate. A series of studies on cellular growth strategies and metabolic network organization [1, 2] has shown that evolutionary and biochemical constraints shape the set of feasible cellular states in a highly structured way. In metabolic network analysis, stoichiometric constraints restrict the space of possible metabolic fluxes to a convex polytope whose extreme rays correspond to elementary metabolic modes [3]. These modes represent fundamental pathways through the network and provide a geometric description of how cells can organize metabolic activity while maintaining biochemical feasibility. This geometric viewpoint suggests that cellular behavior can be interpreted as exploration of structured regions of a high-dimensional space rather than arbitrary movement through expression or flux coordinates.

In developmental biology, however, a different type of structure emerges. Instead of convex feasible regions [4], cellular differentiation often follows branching trajectories in gene expression space. Cells evolve through a sequence of intermediate states that form lineage trees rather than convex geometries. Recovering these lineage relationships from high-dimensional measurements, such as single-cell transcriptomic data, therefore appear as a central problem in modern computational biology.

Dimensionality reduction techniques such as Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) are widely used to visualize such high-dimensional data. While these methods preserve certain geometric relationships between cell states, they do not explicitly recover ancestral structure. Methods designed for trajectory inference aim instead to reconstruct the underlying branching topology of cellular development.

The recently proposed Bonsai algorithm [5] addresses this challenge by reconstructing lineage trees directly from high-dimensional data, using statistical structure in the data to infer

ancestral relationships between cell states rather than relying solely on geometric proximity in low-dimensional embeddings.

2 Objectives

The goal of this small project is to explore the Bonsai lineage inference method in a controlled synthetic setting and to compare the inferred lineage structure with geometric embeddings obtained by PCA and UMAP.

In addition to demonstrating the conceptual distinction between geometric embeddings and lineage topology, the project also aims to gain practical familiarity with the Bonsai software implementation. This includes exploring the algorithmic workflow, evaluating the usability of the codebase, and assessing the computational requirements for applying the method to small example datasets. Such practical considerations are relevant for integrating computational tools into biological research workflows.

3 Methods

3.1 Synthetic branching dataset

To study the behavior of lineage inference methods in a controlled setting, a synthetic dataset representing a simple branching developmental trajectory was generated. Two branches were constructed from a common origin and embedded into a higher-dimensional feature space. Random noise was added to additional dimensions to mimic measurement variability typically present in biological data. The resulting dataset therefore contains a known ground-truth branching structure while remaining embedded in a moderately high-dimensional space.

3.2 Geometric embeddings

To visualize the structure of the synthetic dataset, two commonly used dimensionality reduction techniques were applied. Principal Component Analysis (PCA) [7] provides a linear projection of the data that preserves maximal variance, while Uniform Manifold Approximation and Projection (UMAP) [6] constructs a nonlinear embedding designed to preserve local neighborhood relationships. These embeddings allow visualization of the dataset in two dimensions while approximately preserving geometric relationships between points.

3.3 Lineage inference with Bonsai

Lineage topology was inferred using the Bonsai algorithm. Bonsai reconstructs a tree structure from high-dimensional measurements by identifying ancestral relationships between cell states and organizing them into a lineage graph. Unlike geometric embeddings, which primarily preserve pairwise distances or neighborhood relationships, Bonsai aims to recover the underlying branching topology of cellular trajectories.

In this study, the Bonsai pipeline was executed on the synthetic dataset using the reference implementation provided by the authors. The resulting tree structure was exported in Newick format and processed using standard phylogenetic tools to obtain the lineage graph.

3.4 Evaluation and visualization

To compare geometric embeddings with lineage inference, the reconstructed Bonsai tree was visualized alongside the PCA and UMAP projections. The inferred lineage structure was projected onto the embedding spaces to illustrate how topological relationships inferred by Bonsai relate to geometric proximity in the embeddings.

Quantitative comparison of pairwise distance preservation was considered but not emphasized, as the objectives of geometric embeddings and lineage inference differ fundamentally. PCA and UMAP aim to preserve geometric relationships between points, whereas Bonsai reconstructs branching topology rather than distances in feature space. The comparison therefore focuses primarily on qualitative visualization of the inferred lineage structure.

4 Results

Figure 1 summarizes the comparison between geometric embeddings and lineage inference on the synthetic branching dataset.

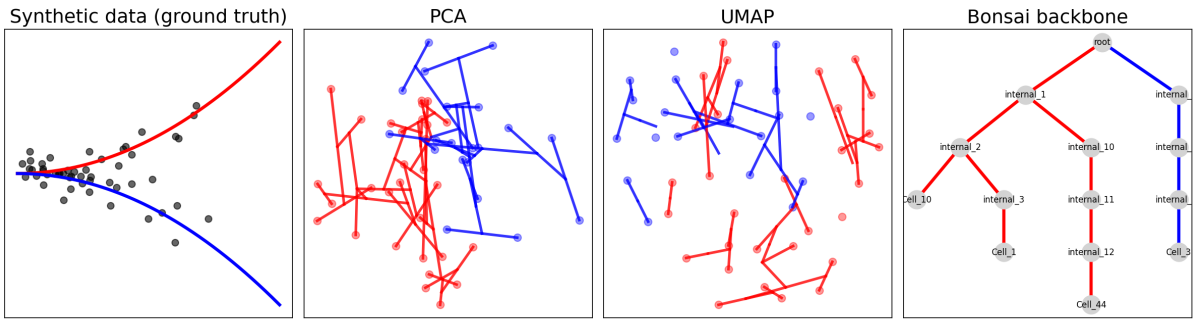

Figure 1: Comparison between geometric embeddings and lineage topology reconstruction for a synthetic branching trajectory. Left: synthetic dataset with two developmental branches originating from a common root. Center: PCA and UMAP embeddings of the same dataset. Colored edges indicate lineage relationships inferred by the Bonsai algorithm and projected onto the embeddings. Right: backbone representation of the inferred Bonsai lineage tree. The figure illustrates the conceptual difference between geometric embeddings, which preserve spatial relationships between observations, and lineage inference methods such as Bonsai, which reconstruct ancestral topology in high-dimensional state space.

The first panel illustrates the synthetic data used in this study. Two branching trajectories originate from a common root and diverge as cells progress along the developmental axis. This structure provides a simple ground-truth model of a branching lineage system embedded in a higher-dimensional feature space.

The second and third panels show two-dimensional embeddings obtained using PCA and UMAP, respectively. In the PCA projection, the two main branches are already reasonably separated along the principal components, reflecting the global variance structure of the dataset. UMAP further organizes the cells according to local neighborhood relationships and produces a visually clear separation of the two trajectories.

For clarity, extremely long projected edges of the Bonsai tree were omitted from the visualization. In the PCA and UMAP panels, line segments represent inferred lineage connections between cell states. Segments terminating at a node correspond to connections leading to terminal cell states, while segments terminating at another segment indicate branching through intermediate (internal) lineage states inferred by the algorithm.

The Bonsai lineage inference results are visualized in two complementary ways. In the PCA and UMAP panels, edges of the inferred Bonsai tree are projected onto the embeddings, connecting cells that Bonsai identifies as belonging to the same lineage path. These connections reveal the inferred developmental trajectories within the geometric structure of the embeddings. Notably, cells connected by Bonsai are not always the closest points in the embedding space, reflecting the difference between geometric proximity and ancestral relationships.

The fourth panel shows the backbone structure of the inferred Bonsai lineage tree. For clarity,

only the major lineage branches are displayed, while shorter side branches representing minor evolutionary dead-ends are omitted. This simplified representation highlights the dominant developmental trajectories inferred from the data.

Branches are colored consistently with the corresponding trajectories in the embedding plots, allowing direct comparison between the geometric and topological representations of the data. Together, these visualizations illustrate how lineage topology reconstructed by Bonsai relates to the geometric structure captured by dimensionality reduction methods.

5 Discussion

This small demonstrative study highlights the conceptual distinction between geometric embeddings and lineage topology in high-dimensional biological data. Dimensionality reduction methods such as PCA and UMAP provide useful visualizations of cellular state spaces by preserving certain geometric relationships between observations. However, these methods do not explicitly reconstruct ancestral relationships between cell states.

The Bonsai algorithm approaches the problem from a different perspective. Rather than focusing on geometric proximity in a low-dimensional embedding, Bonsai attempts to reconstruct the branching topology of developmental trajectories directly from the high-dimensional measurements. As illustrated in the visualizations, cells that Bonsai connects within the same lineage path are not always the nearest neighbors in the PCA or UMAP projections. This reflects the fundamental difference between geometric similarity and evolutionary or developmental ancestry.

From a conceptual perspective, the results support the broader view that biological systems often explore structured regions of high-dimensional state spaces. In metabolic systems, stoichiometric constraints define convex polytopes that describe feasible flux configurations. In developmental systems, the structure instead takes the form of branching lineage trajectories through gene expression space. Methods such as Bonsai provide computational tools to recover this topological structure from experimental measurements.

From a practical perspective, this project also provided an opportunity to explore the Bonsai software implementation. The algorithm can be applied to synthetic datasets using the publicly available reference implementation, although the workflow requires explicit configuration and intermediate file handling. Nevertheless, the resulting lineage reconstructions can be readily analyzed using standard phylogenetic representations and graph-based tools.

Future work could extend this exploration by applying Bonsai to larger simulated differentiation systems or to publicly available single-cell datasets. In addition, systematic evaluation metrics for lineage reconstruction could be explored to complement the qualitative visualization presented here.

Such approaches illustrate how computational methods can complement geometric data visualization by revealing the underlying generative structure of biological state spaces. Developing and evaluating tools that connect geometric representations with lineage topology will remain an important component of computational analysis in moder1n single-cell biology.

6 Reproducibility

All experiments were implemented in Python using NumPy, scikit-learn, UMAP, and the Bonsai reference implementation.

All code used to generate the synthetic datasets, run the Bonsai pipeline, and produce the visualizations is available in the accompanying repository:

`https://github.com/franzm64/bonsai-lineage-demo`

References

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